

The background of the image is a solid dark blue. Overlaid on this background is a large, faint, circular pattern composed of numerous small red dots. These dots are arranged in a way that they form a ring around the central text, with some dots also scattered within the ring. The dots are more densely packed in some areas, creating a sense of depth and movement.

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Multiscale and Multilayer Contrastive Learning for Domain Generalization

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1. Introduction
2. The $M^2 - CL$ Architecture
3. Experimental
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The Core Objective

Train a parametric model $f(\cdot; \theta)$ on a set of N available **source domains** $\mathcal{S} = \{S_1, S_2, \dots, S_N\}$ such that it generalizes robustly to K completely **unseen target domains** $\mathcal{T} = \{T_1, T_2, \dots, T_K\}$.

Why Is This Hard?

- ▶ **Zero access** to target-domain data during training — no direct supervision signal.
- ▶ Models must learn **domain-invariant** features from source data alone.
- ▶ Distribution shift at test time causes catastrophic accuracy drops in naive ERM.

Representative DG Approaches

- ▶ **ERM:** Aggregates all source data — learns domain-specific shortcuts.
- ▶ **CORAL / MMD:** Aligns first/second-order statistics — **single layer only**, loses geometric detail.
- ▶ **RSC:** Drops dominant features at training — ignores multi-scale structure.
- ▶ **Mixup:** Interpolates images across domains — does not enforce feature invariance.

What M^2 -CL Contributes

- ▶ **Multi-scale & multi-layer** extraction — captures geometry at every depth.
- ▶ **Spatial Dropout** in each block — prevents style shortcuts from leaking.
- ▶ **Layer-wise contrastive loss** — enforces same-class alignment across domains at *all* scales simultaneously.

Why ERM Fails: Shortcut Learning

The ERM Trap

- ▶ Minimizes training loss by exploiting **superficial shortcuts** (textures, backgrounds) rather than object geometry.
- ▶ These cues **vanish or invert** under domain shift.

This Motivates M^2 -CL

- ▶ Need to **force the model** to rely on structural, domain-invariant features.

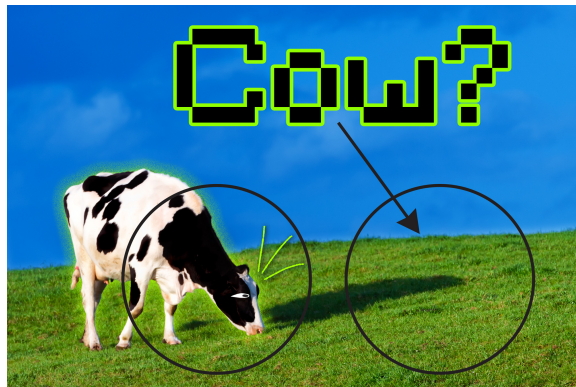


Fig: ERM relies on “Grass” (shortcut) instead of actual “Cow” geometry.

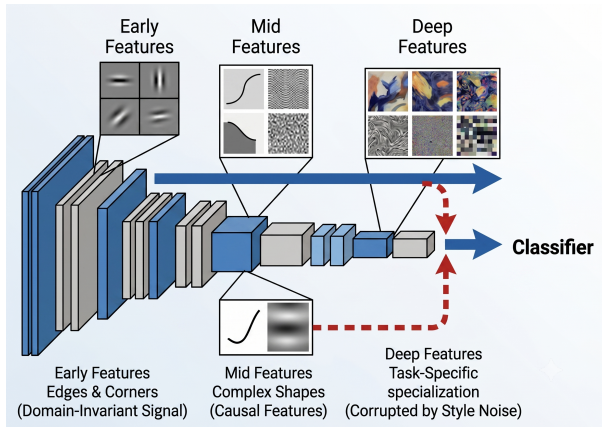
The Last-Layer Problem

- ▶ Standard CNNs use **deepest features** only.

The Invariant Signal

- ▶ **Early/Mid Layers:** Hold shapes and edges.

⇒ Corrupted by style and filters out raw structure details.



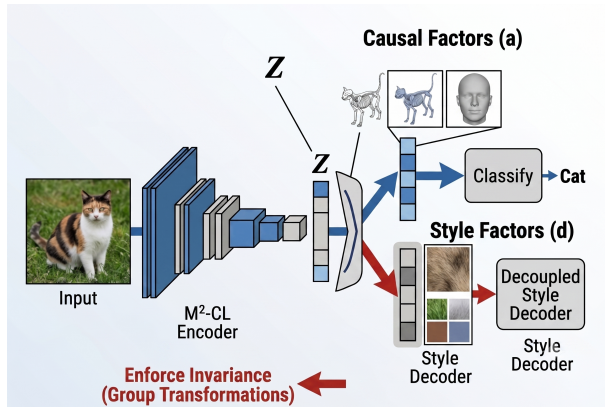
Causal vs. Style Attribute Disentanglement

Latent Space Split: $X \approx (a, d)$

- ▶ **Causal (a):** Class-specific geometry (Invariant).
- ▶ **Style (d):** Domain-specific context and texture (Noise).

The DG Goal

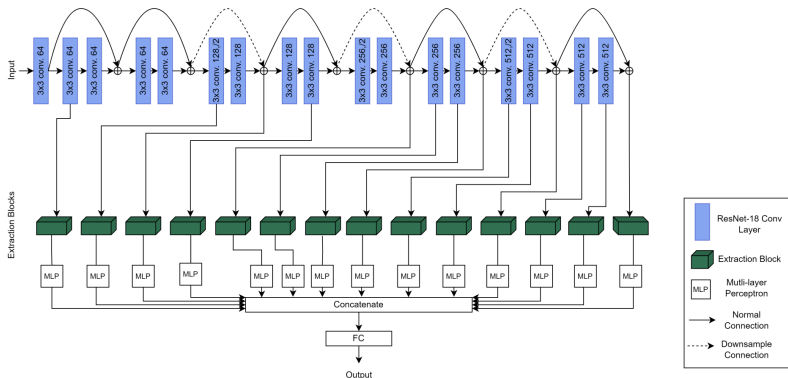
Isolate **Causal** a from **Style** d to ensure the model only relies on invariant geometry for prediction.

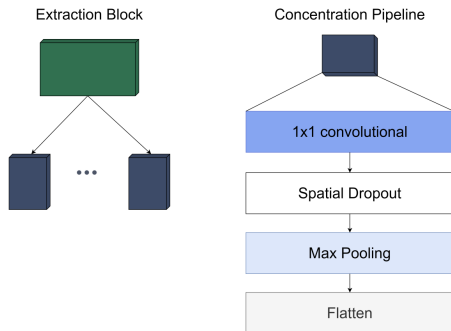


The M^2 Architectural Paradigm

Framework Concept

Leverages parallel multi-scale concentration blocks mapped across multiple intermediate layers of a standard backbone network.





Concentration pipeline layout.

Internal Sequence Mechanics

1. **1×1 Convolutional Layer:** Compresses high-dimensional channels c down to $\lfloor c/r \rfloor$ using a reduction factor r .
2. **Spatial Dropout Layer:** Mitigates pixel correlation leakage.
3. **Multi-Scale Max Pooling:** Captures spatial features before reduction.
4. **MLP Projection Vector:** Maps pooled features into vectors via MLPs.

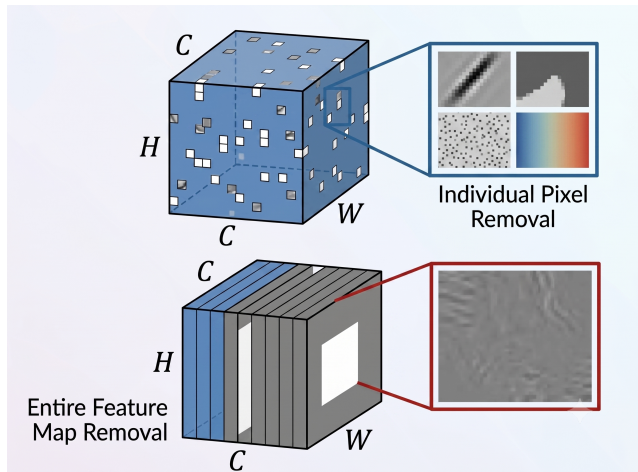
Spatial Dropout

Vanilla Dropout

Drops **individual pixels** — adjacent pixels remain correlated; style information **leaks through**.

Spatial Dropout

Drops **entire feature maps** ($H \times W$ channels). Forces the network to rely on alternative feature maps, **blocking style shortcuts** at the source.



Multi-Scale Concentration & Hypercolumn Construction

Spatial Scaling Allocations

- ▶ **Early Hooks:** Pooling sizes optimized for 8×8 , 4×4 , and 2×2 .
- ▶ **Late Hooks:** Deeper fields adapt to 7×7 and 3×3 dimensions.

Concatenation Protocol

Flattened multi-scale vectors are concatenated into a **hypercolumn vector** for final FC layer classification.

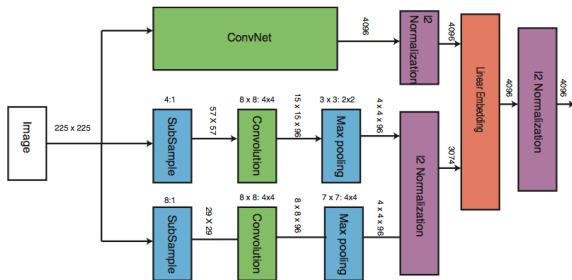


Fig: Multi-scale pooling pipelines merging into a unified Hypercolumn vector.

Geometric Constraints

- ▶ **Pull:** Same class, cross domains.
- ▶ **Push:** Different classes, same style.

Matrix Parallelization

- ▶ Prevents memory bottlenecks.
- ▶ Stacks batch: $U^{(l)T} U^{(l)}$.
- ▶ Denominator computed **once**.

Alignment Measure ($\|u\| = 1$)

$$p^{(l)}(c) = \frac{\sum_{i,j \in N_c} \exp(u_i^T u_j / \tau)}{\sum_{i,a \in N_b} \exp(u_i^T u_a / \tau)}$$

Total Objective

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \sum_{l=1}^M \mathcal{L}^{(l)}$$

$$\mathcal{L}^{(l)} = - \sum_{c=1}^C \log p^{(l)}(c)$$

Optimizer

- ▶ **SGD, 30 epochs:** Stable convergence; training curves plateau before epoch 30.
- ▶ **LR = 0.001:** Standard for fine-tuning pretrained ResNet; avoids over-shooting.
- ▶ **Batch = 128:** Contrastive loss requires enough positive/negative pairs per step.

Architecture & Loss

- ▶ $r = 4$: Ablation (Table 3) shows best accuracy/memory trade-off; $r=2$ overfits, $r=6$ under-compresses.
- ▶ $\alpha = 0.01$: Contrastive term guides without dominating CE loss; larger α collapses to 64% PACS.
- ▶ $\tau = 1.0$: Robust default; sensitivity analysis (Appendix) shows stable performance across $\tau \in [0.8, 1.4]$.



VLCS

10,729 images | 5 classes

Domains: PASCAL, LabelMe, Caltech, SUN



(a) VLCS

PACS

9,991 images | 7 classes

Domains: Photo, Art, Cartoon, Sketch



(b) PACS

Office-Home

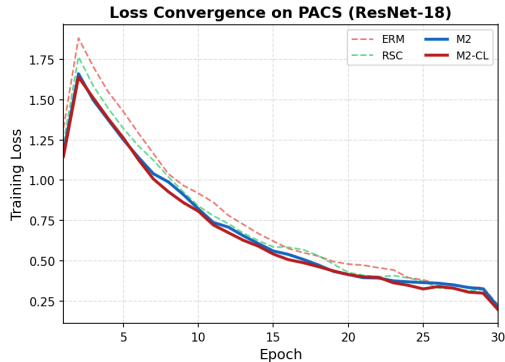
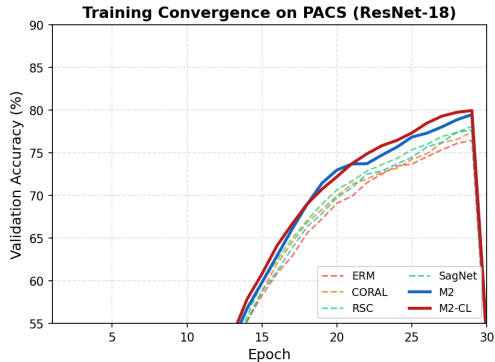
15,588 images | 65 classes

Domains: Art, Clipart, Product, Real-World



(c) Office-Home

Training Convergence on PACS (ResNet-18)



M^2 -CL converges faster and plateaus **higher** than all baselines. The contrastive regularizer suppresses loss spikes caused by domain-conflicting gradients.

Results: PACS and VLCS (ResNet-18)

Method	Art	Cartoon	Photo	Sketch	PACS Avg	Caltech	Labelme	Sun	Voc	VLCS Avg
ERM	76.53	74.80	94.95	67.89	78.54	96.87	65.03	71.62	71.60	76.28
RSC	79.26	76.12	95.07	70.88	80.34	97.68	66.33	73.53	72.75	77.57
CORAL	78.55	72.74	92.24	73.93	79.36	95.16	60.73	70.66	65.00	72.89
MIXUP	79.79	72.81	93.11	63.97	77.42	93.44	60.25	74.36	69.87	74.48
MMD	77.75	73.35	92.34	69.74	78.29	97.28	60.62	70.48	65.21	73.40
SagNet	78.71	74.90	94.79	70.48	79.72	95.76	61.22	69.61	71.90	74.62
SelfReg	79.46	76.32	95.93	71.60	80.83	92.53	62.56	70.27	72.15	74.38
ARM	79.77	75.33	94.43	<u>74.76</u>	81.07	95.86	60.19	70.66	72.83	74.89
EQRM	78.61	72.38	92.48	<u>71.13</u>	78.65	96.17	60.57	70.44	64.66	72.96
SAGM	78.22	75.14	94.79	70.46	79.65	96.37	66.22	71.14	71.35	76.27
M2	<u>80.64</u>	<u>76.91</u>	95.50	74.20	<u>81.81</u>	<u>98.24</u>	<u>67.02</u>	73.80	72.92	<u>78.00</u>
M2-CL	81.04	77.48	<u>95.70</u>	75.81	82.51	98.69	67.48	<u>74.05</u>	73.76	78.50

Dataset Table : Office-Home (ResNet-18)

Top-1 accuracy on Office-Home, ResNet-18.

Method	Art	Clipart	Product	Real-World	Avg
ERM	56.12	47.88	<u>72.47</u>	73.58	62.51
RSC	56.00	46.09	<u>70.85</u>	72.32	61.31
CORAL	55.46	49.39	71.75	73.28	62.47
MIXUP	56.37	<u>49.81</u>	70.87	73.61	62.66
MMD	55.67	48.52	71.68	73.19	62.27
SagNet	55.75	47.74	71.71	73.17	62.09
SelfReg	57.56	49.69	72.43	73.72	63.35
ARM	54.47	48.45	71.14	<u>73.79</u>	61.96
EQRM	56.08	48.34	71.57	72.94	62.23
SAGM	55.79	46.51	71.16	73.15	61.65
M2	<u>58.04</u>	49.30	73.04	73.50	<u>63.47</u>
M2-CL	58.77	50.86	72.20	74.91	64.18

Average Accuracy

- ▶ **ResNet-18:** M^2 -CL reaches **82.51** on PACS, **78.50** on VLCS, and **64.18** on Office-Home.
- ▶ The average score improves over the strongest baseline in each benchmark summary.

Interpretation

- ▶ M^2 improves feature representations through multi-scale reuse.
- ▶ M^2 -CL aligns same-class features across domains.
- ▶ Ablation studies identify the sources of these gains.

Ablation Study on PACS and VLCS

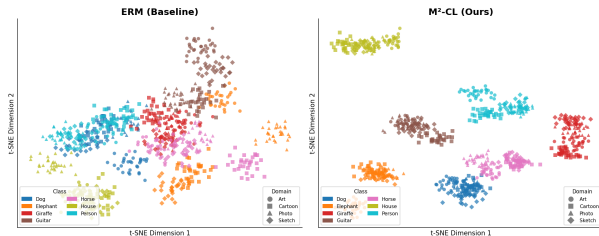
pipe	r	drop	loss	A	C	P	S	PACS Avg	C	L	S	V	VLCS Avg
c	2	-	-	72.34	72.29	94.19	69.51	78.23	93.87	<u>60.91</u>	65.89	61.91	71.22
c	4	-	-	74.38	72.74	90.13	69.71	77.21	94.14	<u>59.01</u>	67.82	65.42	69.62
c	6	-	-	75.72	70.12	94.26	73.33	77.92	91.93	56.81	67.28	64.10	68.95
c	2	yes	-	75.67	73.29	92.16	<u>75.95</u>	77.93	93.73	57.72	66.34	62.50	71.74
c	4	yes	-	74.36	74.05	92.57	76.45	79.37	<u>96.07</u>	56.28	67.07	64.30	65.81
c	6	yes	-	75.97	72.77	93.01	74.73	76.81	<u>92.94</u>	59.73	66.15	64.51	70.09
p	2	-	-	76.09	55.88	68.35	60.92	64.40	95.60	56.97	66.25	73.15	<u>73.96</u>
p	4	-	-	74.72	74.62	92.01	70.32	77.64	95.20	58.10	66.20	64.90	<u>70.29</u>
p	6	-	-	75.38	69.99	91.21	69.47	77.77	95.23	56.81	69.13	65.34	70.55
p	2	yes	-	74.61	72.39	87.08	73.95	78.88	95.73	58.33	68.44	63.74	70.49
p	6	yes	-	<u>76.73</u>	73.98	93.41	73.56	<u>79.84</u>	93.19	56.05	66.62	64.24	70.43
p	4	yes	-	<u>76.39</u>	75.98	<u>95.58</u>	74.49	<u>79.15</u>	94.54	59.77	67.26	<u>74.55</u>	73.14
p	4	yes	yes	80.43	<u>75.43</u>	96.87	74.42	80.10	96.09	61.15	<u>69.05</u>	75.70	74.15

Table: Reference Insights (PACS / VLCS, ResNet-18)

- ▶ **Best full configuration:** $p, r = 4$, $drop = yes$, $loss = yes$ reaches **80.10** PACS Avg and **74.15** VLCS Avg.
- ▶ **Contrastive loss contribution:** adding the loss to the same $p, r = 4$, $drop = yes$ setting improves PACS from **79.15** to **80.10** and VLCS from **73.14** to **74.15**.
- ▶ **Spatial dropout effect:** for $p, r = 4$, dropout improves PACS from **77.64** to **79.15** and VLCS from **70.29** to **73.14**.

Feature Space: t-SNE Visualization

Feature Space Visualization — PACS (Sketch target domain, ResNet-18)



ERM (Left)

Same-class points from different domains are **scattered** — the model encodes domain identity, not class identity.

M²-CL (Right)

The layer-wise contrastive loss **collapses intra-class domain scatter**: all domains for the same class cluster tightly, while inter-class boundaries stay sharp.

Insight: The network has learned truly *domain-invariant* representations, directly validating the contrastive objective.

Saliency Maps: What the Model Focuses On

PREDICTION

Volcano

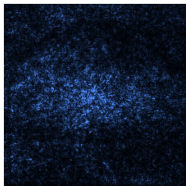
Confidence 100.0%

Volcano	100.0%
Torch	0.0%
Tent	0.0%
Yurt	0.0%
Alp	0.0%

Input Image

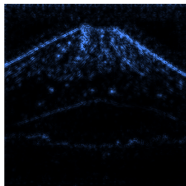


ERM (Baseline)



Scattered attention · 61.1% of pixels active

M²-CL (Ours)



Focused attention · 19.7% of pixels active

Bright blue pixels = high gradient magnitude · Dark = model ignores

Guided backprop (M²-CL) is 3.1x more spatially concentrated than vanilla gradient (ERM)

ERM

Saliency spreads across **backgrounds and textures** — the model fires on domain-specific context (grass, sky) rather than object structure.

M²-CL

Saliency concentrates on **object contours and edges**, consistent across Photo, Art, Cartoon, and Sketch domains.

Insight: The contrastive loss forces the network to anchor decisions on *structural geometry*, not style — the intended inductive bias.

Core Achievements

- ▶ **Architecture:** Successfully combines multi-scale, multi-layer intermediate features without propagating domain-specific styles through the network.
- ▶ **Loss Optimization:** Introduces a layer-wise contrastive loss that enforces domain invariance across all extraction levels.
- ▶ **Empirical Validation:** Outperforms previous domain generalization methods and sets a new state-of-the-art across four major benchmark datasets.

Current Computational Drawbacks

- ▶ **Memory Overhead:** Concatenating multiple intermediate feature maps creates dense hypercolumns before the classification head, increasing memory usage.
- ▶ **Batch Size Dependency:** Like most contrastive methods, M^2 -CL requires large training batches to construct effective positive/negative pairs.

Future Research Avenues

Explore cross-layer attention mechanics, build causal inference systems, and evaluate alternative similarity metrics like KL divergence to refine latent distribution shapes.

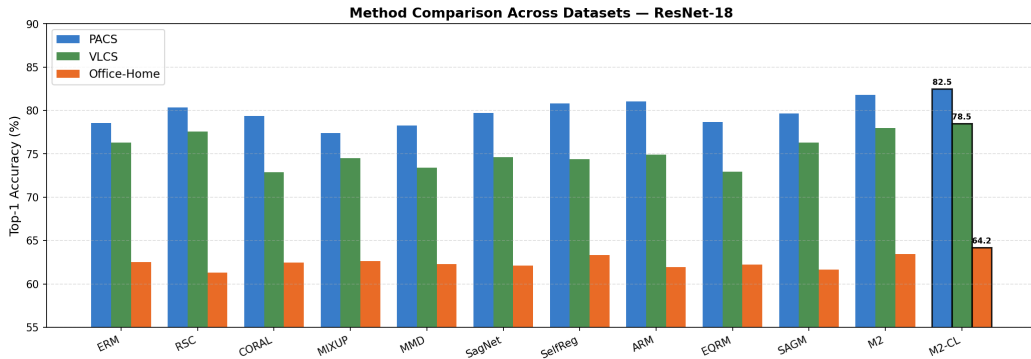
A decorative graphic on the left side of the slide. It features a dark blue background with a large, stylized circular pattern composed of many small red dots. The dots are arranged in concentric, slightly offset rings, creating a sense of depth and movement. The word "HUST" is centered within this pattern.

HUST

THANK YOU !



Method Comparison — All Datasets (ResNet-18)



Dataset Table: PACS and VLCS (ResNet-50)

Top-1 accuracy on PACS and VLCS, ResNet-50.

Method	Art	Cartoon	Photo	Sketch	Avg	Caltech	Labelme	Sun	Voc	Avg
ERM	83.54	78.47	96.51	73.64	83.04	97.01	62.22	71.79	76.09	76.78
RSC	82.08	80.26	96.51	75.45	83.57	97.81	64.44	74.60	75.53	78.10
CORAL	83.94	79.27	94.37	81.98	84.89	97.01	61.34	75.26	71.74	76.34
MIXUP	85.16	77.40	96.23	78.26	84.26	96.94	62.33	73.40	72.32	76.25
MMD	82.21	74.70	92.89	69.00	79.70	97.48	63.85	75.49	71.65	77.12
SagNet	85.23	78.47	93.20	77.71	83.65	94.14	60.43	70.49	69.96	73.75
SelfReg	83.33	79.06	93.16	77.24	83.20	92.49	60.73	66.80	73.58	73.40
ARM	82.14	79.10	92.97	73.02	81.81	96.01	63.98	66.26	71.79	74.51
EQRM	84.98	77.74	91.26	74.88	82.21	94.35	60.80	66.35	72.94	73.61
SAGM	81.35	78.99	93.24	79.39	83.24	97.39	64.23	69.30	76.17	76.77
M2	<u>86.02</u>	79.90	<u>97.03</u>	81.50	<u>86.11</u>	97.52	<u>64.77</u>	<u>76.02</u>	<u>76.72</u>	<u>78.76</u>
M2-CL	86.48	<u>80.05</u>	97.99	<u>81.70</u>	86.56	<u>97.60</u>	64.86	76.65	77.26	79.09

Dataset Table: Office-Home (ResNet-50)

Top-1 accuracy on Office-Home, ResNet-50.

Method	Art	Clipart	Product	Real-World	Avg
ERM	63.43	51.32	76.45	<u>78.13</u>	67.33
RSC	62.64	51.55	75.11	<u>77.34</u>	66.66
CORAL	61.27	<u>54.34</u>	73.35	77.83	66.70
MIXUP	63.23	51.91	<u>77.10</u>	76.39	67.16
MMD	59.63	49.37	75.28	74.71	64.75
SagNet	61.41	51.85	72.85	73.79	64.97
SelfReg	59.95	51.22	71.28	74.52	64.24
ARM	54.23	52.14	71.73	71.26	62.34
EQRM	63.52	52.32	73.28	76.45	66.39
SAGM	61.01	53.27	74.81	77.00	66.52
M2	<u>65.35</u>	53.90	76.80	78.74	<u>68.70</u>
M2-CL	66.46	54.82	78.24	77.85	69.34

Table: Sensitivity to Temperature τ

Sensitivity analysis for τ in M2-CL, ResNet-18.

τ	PACS	VLCS
0.01	78.10	76.15
0.1	80.03	77.27
0.2	81.37	74.71
0.4	82.07	72.03
0.6	83.70	74.15
0.8	84.67	75.21
1.0	80.07	76.24
1.2	81.34	75.11
1.4	83.14	75.69
1.6	82.17	76.84
1.8	80.51	74.66
2.0	78.72	74.99
10.0	79.92	74.04
100.0	81.84	73.74

Table: Sensitivity to Loss Weight α

Sensitivity analysis for α in M2-CL, ResNet-18.

α	PACS	VLCS
0.00	82.76	72.36
10e-5	80.41	73.56
10e-4	81.17	73.71
10e-3	82.29	74.38
10e-2	79.70	73.39
10e-1	64.04	55.96

Statistical Alignment

- ▶ **CORAL:** Aligns second-order statistics (covariance) across source domains.
- ▶ **MMD:** Minimizes Maximum Mean Discrepancy between domain distributions.

Data Augmentation / Interpolation

- ▶ **Mixup:** Interpolates images and labels across domain pairs.
- ▶ **SagNet:** Style-adversarial training to separate content from style.

Feature Regularization

- ▶ **RSC:** Randomly suppresses dominant activations to prevent shortcut reliance.
- ▶ **SelfReg:** Self-regularization via contrastive and distance-based losses.
- ▶ **ARM:** Context-conditional adaptation at inference time.
- ▶ **SAGM:** Sharpness-aware gradient matching for flat loss minima.
- ▶ **EQRM:** Empirical quantile risk minimization to reduce tail risk.

Tables 6–7 Insights: Hyperparameter Tuning

- ▶ **Temperature optimum:** PACS peaks at **84.67** with $\tau = 0.8$, while VLCS peaks at **77.27** with $\tau = 0.1$.
- ▶ **Default-scale behavior:** $\tau = 1.0$ remains competitive with **80.07** PACS and **76.24** VLCS.
- ▶ **Loss-weight sensitivity:** $\alpha = 10e^{-3}$ gives the best VLCS score, **74.38**, while $\alpha = 0$ gives the highest PACS score, **82.76**.

Alpha Bounds Limits

$$\alpha = 10e^{-1} \implies \text{PACS } 64.04, \quad \text{VLCS } 55.96$$

A large contrastive weight dominates cross-entropy optimization and causes a sharp drop compared with the best scores in Table 7.